

5.3.3 Kidney functions and malfunctions

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the need for the removal of excretory products	To include the reasons for the deamination of amino acids and production of urea in liver cells (no details of the ornithine cycle are required).
(b) the structure of the kidney as part of the excretory system	To include the gross structure of the kidney (cortex, medulla, renal pyramids, renal pelvis and ureter) and relative positions of the major blood vessels and organs associated with the kidney (renal artery and vein, bladder and urethra). PAG2
(c) (i) the structure and function of the kidney nephron related to the processes of ultrafiltration and selective re-absorption which result in the production of urine	To include the use of microscopy, diagrams and photomicrographs to examine and identify structures. <i>M0.1, M0.2, M0.3, M1.8, M2.2</i> PAG1 HSW8
(ii) practical investigations into the biochemical composition of 'mock' urine, renal artery and renal vein plasma and filtrate	PAG9 HSW3, HSW4, HSW5, HSW6
(d) the role of the kidney in osmoregulation	To include the location and role of osmoreceptors, the secretion of ADH from the posterior pituitary, the action on ADH at the collecting ducts, the role of cyclic AMP in collecting duct cells and the insertion and removal of aquaporins into cell surface membranes.
(e) the role of the kidney as an endocrine gland	To include an outline of the homeostatic function of erythropoietin (EPO) and renin (angiotensin).
(f) the causes, diagnosis and consequence of kidney failure	To include causes (uncontrolled diabetes, kidney stones, blood pressure and bacterial infections), diagnosis (analysis of data from laboratory samples e.g. urine or blood composition) and consequences (renin and EPO changes and cardiovascular disease). <i>M0.1, M0.3, M2.2, M2.3, M2.4</i> PAG9 HSW9, HSW10, HSW11, HSW12

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| (g) the use of haemodialysis and peritoneal dialysis in the treatment of kidney failure | To include a consideration of the advantages and disadvantages of both forms.
<i>M1.4</i>
HSW9, HSW10, HSW11, HSW12 |
| (h) the use of transplant surgery in the treatment of kidney failure | To include practical issues involved in the use of donor organs for kidney transplants.
HSW7 |
| (i) the future for transplant surgery. | To include the implications of therapeutic cloning and reproductive cloning and the use of stem cell technology.
HSW7 |